

Single molecular rotations/reorientations

The contributions of molecular rotations/reorientations can be treated by the model developed by Nordio (25). Here,

$$(T_1^{-1})_R(\omega_L) = \frac{3}{2} K_{dd}^2 I(I+1) \left[\mathcal{J}_{\mathcal{R}}^{(1)}(\omega_L) + \mathcal{J}_{\mathcal{R}}^{(2)}(2\omega_L) \right] \quad (\text{S2})$$

where the rotational correlation functions used to get spectral densities $\mathcal{J}(\omega)$ are here expressed for cylindrical molecules reorienting in uniaxial mesophases. Following Abragam's notation (26) ($m_L \neq 0$):

$$\mathcal{J}_{\mathcal{R}}^{(m_L)}(m_L \omega_L) = \frac{4}{3} (m_L)^2 \sum_{m_M=0}^2 A^{(m_M)} c(m_L, m_M) \frac{(\tau_{m_L m_M}^2)^{-1}}{m_L^2 \omega_L^2 + (\tau_{m_L m_M}^2)^{-2}} \quad (\text{S3})$$

with the correlation times $\tau_{m_L m_M}^2$ given by

$$(\tau_{m_L m_M}^2)^{-1} = \frac{\tau_S^{-1}}{\beta_{m_L m_M}^2} + (\tau_L^{-1} - \tau_S^{-1}) m_{m_H}^2 \quad (\text{S4})$$

while $c(m_L, m_M)$ and $\beta_{m_L m_M}^2$ can be estimated numerically as functions of the nematic order parameter S . The average molecular geometrical factors $A^{(0)} = \overline{\sum_{j \neq i} |d_{0,0}^2(\alpha_{kl})|^2} / r_{kl}^6$ and $A^{(m_M)} = 2 \overline{\sum_{j \neq i} |d_{m_M,0}^2(\alpha_{kl})|^2} / r_{kl}^6$ for $m_M \neq 0$ were estimated for 5CB- α d₂, assuming the most stretched molecular conformation.

In the isotropic phase the nematic order parameter S is zero and in Eq. (3), $c(m_L, m_M) = 1/5$ and $\beta_{m_L m_H}^2 = 1/6$. τ_L^{-1} and τ_S^{-1} are the rotational correlation times along the long and short molecular axis, respectively.

Correlated rotations/reorientations and short-range order fluctuations

$$J^{(k)}(\omega_0) = \frac{4}{3} c_k \sum_{m=-2}^2 \frac{\overline{|d_{m0}^2(\alpha_{ij})|^2}}{r_{ij}^6} \langle |D_{km}^2(\theta)|^2 \rangle \frac{\tau_{km}^2}{1 + \omega_0^2 (\tau_{km}^2)^2} \quad (\text{11.18})$$

$d_{km}^2(\alpha_{ij})$ is the second-rank reduced Wigner rotation matrix and $\langle D_{km}^2(\theta) \rangle = \langle D_{k|m|}^2(\theta) \rangle$ is the second-rank rotation matrix.

$$\frac{\overline{|d_{m0}^2(\alpha_{ij})|^2}}{r_{ij}^6} = \begin{cases} \overline{(3 \cos^2 \alpha_{ij} - 1)^2 / (4 r_{ij}^6)}, m=0 \\ \overline{3 \sin^2 2\alpha_{ij} / (8 r_{ij}^6)}, m=\pm 1 \\ \overline{3 \sin^4 \alpha_{ij} / (8 r_{ij}^6)}, m=\pm 2 \end{cases} \quad (\text{11.19})$$

$$\begin{aligned} & \langle |D_{k|m|}^2(\theta)|^2 \rangle \\ &= \frac{1}{5} + \frac{1}{35} \begin{bmatrix} 10S + 18\langle P_4 \rangle - 35S^2 & 5S - 12\langle P_4 \rangle & -10S + 3\langle P_4 \rangle \\ 5S - 12\langle P_4 \rangle & \frac{5}{2}S + 8\langle P_4 \rangle & -5S - 2\langle P_4 \rangle \\ -10S + 3\langle P_4 \rangle & -5S - 2\langle P_4 \rangle & 10S + \frac{1}{2}\langle P_4 \rangle \end{bmatrix} \quad (\text{11.20}) \end{aligned}$$